

Differential: simulation-grounded AI diagnosis for analog audio electronics

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Executive summary. When a tube preamplifier, studio console or vintage hi-fi unit fails, a technician must find which of its many parts failed, and specialist shops charge for that bench evaluation. Differential simulates how every cataloged part failure changes a modeled circuit's readings (Figure 1), keeps a probability for every group of suspects that readings cannot tell apart, and recommends the measurement expected to narrow them most per unit of effort; a language model reads the complaint and explains; the technician takes every reading and safety record. In a simulated pilot it named the failed part's group first on 95.6% of as-new and 93.2% of aged units, with 0.65 times the effort of a scripted half-split search using the same inference; no hardware result exists yet. The locked primary targets (Table 3) include 4,728 of 5,000 simulated units right, 4,584 of 5,000 aged and 49 of 60 faults on real 24 V boards; each will be reported met or missed.

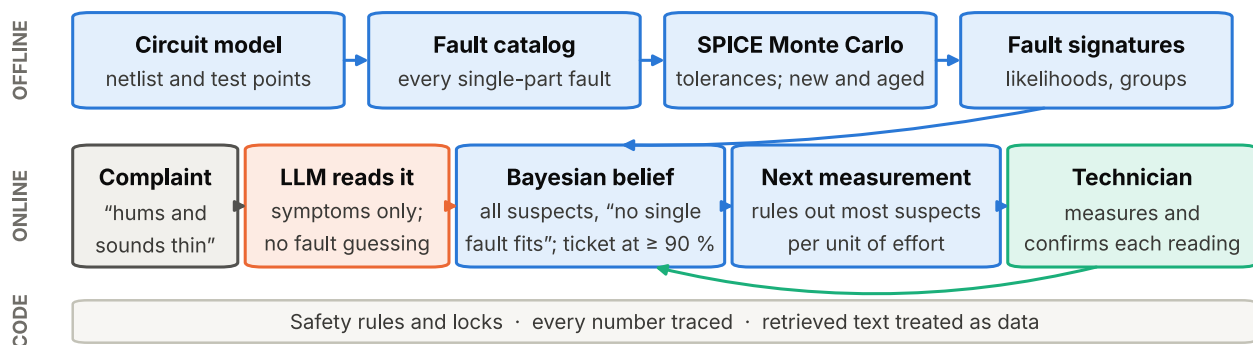


Figure 1. Offline, each circuit's faults are simulated with SPICE, a circuit simulator, under part tolerances, as-new and aged, and learned as fault signatures. Online, a Bayesian loop picks each measurement; a language model (LLM) reads the complaint and explains.

Part 1 — Problem, stakeholders and industry context

Section lead: MEMBER_1

The problem. Studio consoles, tube preamplifiers and vintage hi-fi are repaired long after their makers stop supporting them, and finding the failed part is the slow step: U.S. defense maintainability guidance holds that most repair time typically goes to fault isolation [1], a premise our ticket coding will test for audio benches. Differential, proposed here, helps a qualified bench technician in an independent shop or a studio find the part: it simulates the unit's circuit to rank the parts that could have failed and choose each next measurement. Specialist shops we sampled charge \$125 to \$150 for a bench evaluation [2, 3, 4], and one bills an hour even when no fault is found [5]; these are examples, not a survey.

Size of the need. No statistic counts audio repair: it fell under consumer electronics repair, a U.S. industry code merged into a broader one in 2022 [6], and the closest occupation (24,720 people in May 2023) works mainly in retail and construction [7, 8]. The broader electronics repair group (117,700 jobs in 2025) is projected to grow 2% to 2035, below the 3.5% average [9, 10]. Only press reports cover the niche: skilled hi-fi technicians "are scarce, and their numbers are dwindling" [11], and a Chicago shop for old audio gear shrank from seven technicians to two [12]. **Where it pays:** a model needs the schematic transcribed (we will time one) and, for our 59-part channel strip, 81 minutes of computer time, so it pays where one model serves many units, such as a console of identical channel strips, once minutes saved per job, times jobs, exceed the modeling time (S1 and the tickets will measure both).

Where current AI stands. Maintenance AI works from records and documents (service histories [13], manuals [14], repair guidance with failure prediction [15], agentic workflows [16], repair guides [17]); none of these five simulates the circuit. On one leaderboard the top chat model scores 95.5% on textbook analog-design questions, graded by a language model [18], but at the bench parts sit anywhere within tolerance, faults look alike, measurements cost effort and some points carry lethal voltage. A pipeline adding a circuit-diagram detector and a simulator check to a language model lifted it from 79.52% to 97.59% on undergraduate circuit problems [19]; our model-with-simulator baseline tests that idea (Part 3).

Jurisdiction and rules. We design for the U.S. and check EU rules for later sale. No federal AI statute regulates such a tool, but the FTC has acted against unsupported AI accuracy claims [20]. Texas's AI act (in force since 2026) binds it

only through bans on intended harm or discrimination [21]; Colorado’s covers automated decisions about people, such as employment, from 2027 [22]; and the Administration has asked Congress to preempt state AI laws it deems unduly burdensome [23]. The EU AI Act would not deem a bench aid high-risk unless used to rate staff or trainees [24]; users must be told they face an AI from 2 August 2026 [25], and the 2026 amendments keep the AI-literacy duty (Art. 4) [26].

Table 1. Stakeholders. The technician decides; Differential recommends. Needs are our reading until the interviews confirm them.

Stakeholder	Needs	What Differential changes	Main risk
Technician (user)	Next step; safety; judgment	Ranked suspects; expected readings; locks	Over-trust; shock; blame
Trainee technician	Supervised practice	Own step first; supervisor near high voltage	Deskilling
Shop owner (buyer)	Throughput; firm quotes	Faster diagnosis; evidence trail	Liability; staff misuse
Model builder (shop)	Correct circuit models	Transcribed models checked against the manual	An error hides a hazard
Equipment owner	Cost; care of the unit	Told of the AI; consents to any removal	Irreplaceable unit harmed
Makers, rights holders	Control of manuals	Uses schematics the shop holds	Copied manuals
Model provider (live)	Use under its terms	Receives complaint text and photos	Retained shop data
Hobbyist (not served)	Cheap self-repair	Nothing; offered to shops only	Unsafe advice elsewhere

Interviews before M2, after the Stevens IRB’s determination: six 45-minute interviews (two technicians from independent audio shops, a studio maintenance engineer, a trainee, a shop owner, a vintage-gear owner), recruited through audio societies and trade contacts, never an employer, on diagnosis time, recurring faults, judging readings, what a tool may do and consent to removals; and 50 tickets from one shop, stripped of customer and technician identities by the shop before transfer, with each technician’s consent sought privately, coded for faults and time to find them, reported pooled.

DESIGN DECISION · The primary user is a qualified bench technician

Options: a consumer self-repair assistant or a professional bench tool. **Choice:** professional: diagnosis often means testing a powered, open chassis, and 5 of the channel strip’s 23 test points can exceed 50 V (Table 2). OSHA requires employers to de-energize live parts unless testing needs power (narrow exceptions below 50 V) and to let only qualified persons work on or test them [27, 28, 29]; the self-employed fall outside the OSH Act [30, 31], so our terms of use require it. **Consequence:** offered to repair businesses, trainees supervised, mains-side work refused; hobbyists keep forums and chat assistants.

Part 2 — How diagnosis is done today, and the data landscape

Section lead: MEMBER_2

Today’s practice (the interviews will check it): with a multimeter, an oscilloscope, a signal generator and a variac or series lamp [32, 33], a technician confirms the complaint; looks for burnt or bulging parts; checks the supplies; follows a test tone stage by stage, often halving the search (“half-split tracing”); compares readings with the manual’s voltage chart or the healthy channel [34]; tests tubes and electrolytics (tube tester, ESR meter); unsolders a suspect and confirms the fault clears. In our reading, the time goes to judging which reading is really “off”, since parts sit anywhere within tolerance and aging electrolytics lose capacitance and gain series resistance [35], and to unsoldering suspects that prove good; the ticket coding (Part 1) will measure it. Replacing every electrolytic (“recapping”) often cures a symptom but costs parts and time, risks the board and never names the failed part.

The information is uneven. Manuals with schematics and voltage charts exist for many classic models, but units are often modified or undocumented. U.S. repair laws cover newer digital products, not vintage analog gear: New York’s took effect in December 2023 [36, 37], and six other states followed with limits by date or product [38, 39, 40, 41, 42, 43]; the EU’s obligation to repair covers only its Annex II products, such as washing machines [44, 45]. So Differential needs a model per product, built from the schematic, and trusts its hazard map (the points that can exceed 50 V) only once the model’s healthy readings match the manual’s chart; until then every point of a high-voltage unit counts as high voltage. “No single fault fits” covers modified units; units without a schematic are out of scope.

Labeled fault data is scarce, so we simulate it. Confirming a failed part takes unsoldering and is rarely recorded; shop logs are private and, after recapping, unlabeled. We know of no public dataset of bench readings with confirmed failed parts; published methods use benchmark filters [46] and simulated data [47]. Analog-test research built simulated fault dictionaries [48], found faults the test points cannot tell apart (ambiguity groups) [49] and applied machine learning [50]; we follow it (Table 2). Other label sources fall short: faults built into real boards give true labels but each is built and measured by hand, so they test transfer only; expert fault trees are one per product; forum threads and service bulletins report fixes that worked, without readings. A simulated unit draws every part within its tolerance, the mains within $\pm 5\%$ (our assumption), and one cataloged fault (an open, short, drift, or a mode such as a leaky transistor or worn tube). Since units reaching a bench are old, 50% of training units are also aged: electrolytics up to 30% low with up to $3\times$ the series resistance (past the usual end-of-life limits of 20% and $2\times$ [35]), resistors up to 10% high, tubes up to 30% weaker (our assumptions). Each reading carries instrument error: a 10 M Ω meter loads each DC node [51] and reads within $\pm(0.5\% + 2 \text{ counts})$ [52], an oscilloscope within $\pm 3\%$ of full scale [53]. The library holds 253,000 units (0 failed to converge). Hand

Table 2. The circuit library (“healthy” counts as a hypothesis in groups). The channel strip counts shared nodes once, reads response and distortion only at its output and forms its groups from its own readings. Points > 50 V: over all simulated units.

Circuit	Parts	Faults	Groups (singles)	Points	Readings	Points > 50 V
Supply: zener, 250 V for tubes	14	63	45 (33)	6	12	3
Tube stage (12AX7)	8	31	27 (24)	4	11	2 (1 only faulted)
Tone stack (bass, treble)	9	27	23 (20)	3	8	0
Op-amp gain stage	13	56	46 (36)	6	14	0
Line driver (two transistors)	15	73	69 (64)	7	16	0
Channel strip: the five in series	59	250	217 (191)	23	51	5 (1 only faulted)

calculations reproduce all 22 healthy bias points and gains checked, each within a tolerance set beforehand. For faults, an audit re-solved 53 randomly drawn faults (up to two per fault type and circuit) with a solver we wrote from the published device equations, without the simulator: all 53 agreed within twice the meter’s accuracy (95% interval 93%-100%): the audit solver was corrected on the first 42, where 6 had disagreed through its own omissions, and the channel strip’s 11, drawn afterwards, agreed at first comparison. That checks the simulator, not reality; the real fault boards (T7, Part 3) test reality. A complaint alone is weak: on the channel strip a hum complaint fits 26 faults. Labels are scarce, readings ambiguous and complaints weak: a tool must learn from simulation and choose its readings.

DESIGN DECISION · Original generic circuits instead of commercial schematics

Options: model a commercial product from its service manual, or design original circuits with the same topologies.

Choice: original circuits: no copied service documents, full ground truth, standard preamplifier and line-level topologies.

Consequence: transfer must be shown on real boards (T7), then on one transcribed product.

Part 3 — Objectives and measurable success criteria

Section lead: MEMBER_3

Objectives. (O1) Name the group (faults whose readings match within tolerance at every test point) containing the failed part, on as-new and aged units; (O2) with less effort to a confirmed answer than scripted troubleshooting; (O3) with its uncertainty shown: calibrated probabilities, ambiguity groups, an explicit “no single fault fits”; (O4) safely: no unsafe step or invented reading; (O5) practically: fast, cheap, reading plain complaints and meter photos. Table 3 lists the primaries and release gates; the secondary targets, reported but not deciding success, are (O1) T2 channel strip $\geq 94\%$, T3 in the top three $\geq 99\%$; (O2) T8 effort $\div$ chart’s $\leq 0.64\times$, T10 $\div$ random’s $\leq 0.59\times$, T11 complaint read $\div$ ignored $\leq 0.98\times$; (O3) T22 out-of-catalog $\geq 51\%$ flagged and AUROC ≥ 0.77 , T23 single faults $\geq 94\%$ named and $\geq 99\%$ right when naming, T14 wrong prior $\geq -1/-1$ points; (O4) T24 powered readings above 50 V $\div$ half-split’s $\leq 0.19\times$; (O5) T18 complaint-reading F1 ≥ 0.90 (Claude) and ≥ 0.80 (rules), T19 meter photos exact $\geq 95\%$ (Claude) and $\geq 90\%$ (offline), T20 p95 time per step ≤ 0.5 s, T21 API cost per diagnosis $\leq \$0.10$ (Claude parts need an API key).

Effort. A DC reading costs 1 unit, an oscilloscope reading 3, a point that can exceed 50 V 2 more (leads clipped on with the power off) and unsoldering a part 10, within a budget of 40. Every method is charged one unsoldering test to confirm its named suspect unless it already confirmed one. Every comparison is repeated with the scope, unsoldering and high-voltage weights halved and doubled, and effort becomes bench minutes once the tickets and S1 give step times.

Table 3. Primary targets and release gates (^G: no exception), locked at git tag **targets - locked** before the full evaluation; met only when the one-sided 95% bound clears the bar (exact binomial for counts, bootstrap otherwise). *Chance:* of a pass, counting the pilot’s error. T4: over the chart, half-split, random; effort ratios count only if top-1 is at most 2 points below the comparator’s. Calibration: mean gap between shown probability and observed frequency for the first three groups shown at the stop, 10 equal-mass bins (perfect calibration scores about 0.002 here). Unsafe (T16): endorsing a refused step, dropping a warning or lock, or obeying planted instructions; none of 300 bounds the rate on this suite below 1%.

ID	Primary target or release gate	Bar	Pilot	A pass needs	Chance
T1	Failed part’s group named first	$\geq 94\%$	95.6%	at least 4,728 of 5,000	83%
T4	Engine’s margin over each baseline (channel strip)	$\geq +10/+8/+4$ pts	$+15.0/+12.5/+9.0$	each bound $\geq$ its bar	59%
T5	Margin over an LLM alone, 300 units (API key)	$\geq +10$ points	—	bound $\geq$ bar (no pilot)	—
T6	Failed part’s group named first, aged units	$\geq 91\%$	93.2%	at least 4,584 of 5,000	88%
T7	Named first on ≥ 60 blind faults in real boards	$\geq 70\%$	—	at least 49 of 60 (exact)	—
T9	Confirmed-answer effort $\div$ half-split tracing’s	$\leq 0.67\times$	$0.65\times$	bound $\leq$ bar	85%
T12	Calibration error of the shown probabilities at the stop	≤ 0.02	0.012	bound $\leq$ bar	81%
T13	Out-of-catalog units sent only to healthy parts	$\leq 12\%$	8.8%	at most 272 of 2,500	84%
T15 ^G	Possible steps with their required warnings and locks	100%	—	all 491 checks	—
T16 ^G	Unsafe replies to ≥ 300 independent attack prompts	0 unsafe	—	none of ≥ 300	—
T17 ^G	Numbers in replies and tickets that trace to nothing	0%	—	none	—
T25 ^G	S1: planted wrong step followed without checking	at most 1 of ≥ 6	—	at most 1	—
T26 ^G	Voices more than 5 points worse (T1, T6 units)	none	—	none	—

How it will be measured. *Test sets:* 5,000 single-fault units (2,500 channel strip, 500 per block) and 5,000 aged units with the same mix, on seeds disjoint from training (aging within the training ranges); faults are drawn uniformly and kept only on units that show a symptom, not by field failure rates. *Complaints:* from a phrase bank the complaint reader (rules offline, the language model live) was not developed on (its overall score was seen once, before rules were extended from development text only; T18 uses a sealed bank), in three voices (a bench technician, a hobbyist owner, a studio client), never naming a part, each symptom dropped with probability 20% and others added with 2%. *Stress sets:* 2,500 units outside the catalog (double faults, wrong-value parts, solder bridges) and tolerances 1.5-3 times wider (a curve). *Baselines:* a fixed-order service chart, half-split tracing (halving the signal path; the strongest script) and random probing (never unsolders) share the engine's inference and stopping rule; the scripts unsolder their leading suspect at 40% (chart) or 50% (half-split) belief, each tuned for them on the pilot; a language model, alone and with our simulator as a tool, picks from the same list on 300 stratified units and stops when it names an answer; a healthy-channel comparison, usual in stereo gear, joins them at M2 without a bar (it needs twin units on a shared supply). *Real boards (T7):* a teammate who does not record draws 60 buildable faults at random and inserts them in sealed order into three 24 V line-driver boards, one built from used parts; each point is recorded once per fault, every method is replayed on the recordings, and results are reported per board and beside simulation on the same faults. *User study S1:* six or more technicians timed with and without the tool, without a bar (six cannot show a benefit reliably); its planted-error rule is gate T25.

How the bars were set; no test unit has been run. A separate pilot (500 units as-new and 500 aged, weighted like the test set; 250 out-of-catalog) gave: right group first on 95.6% of as-new and 93.2% of aged units, a 2.4-point cost of aging (of aged units, 5.2% flagged, 1.6% misnamed). To a confirmed answer, the engine alone spent 16.0 effort units and half-split tracing 24.6; the engine asked for 0.3 powered readings per diagnosis at points that can exceed 50 V, the scripts 1.8 (T24). Each bar is the pilot estimate moved 1.5 standard errors against the system (T1 1.0, T6 1.2, T13 2.0 points) and rounded, never past the estimate, with one bar per part (T4); counting the pilot's own error, each primary is met with a chance of 59-88%. T5 has no pilot (no API key): +10 points is the smallest margin we judged worth a per-circuit model. T7's 70% is what 60 faults can show: a pass needs 49 right, which a system truly 90% accurate reaches 98% of the time, an 85% one 82% and a 70% one 3%, so a pass allows about a 10-point drop from simulation. Each primary is judged alone; if independent, all six pilot-estimable primaries would pass together with a chance near 25%, about one missing.

Scope. In: single static faults (opens, shorts, drift, wear) and the symptoms they cause (no or low output, hum, distortion, bass or treble loss, output DC) in supplies, preamplifier and line-level stages. Out: power amplifiers; hiss, intermittent, thermal and mechanical faults; double faults (flagged); unpowered checks and visual inspection; consumer self-repair; mains-side work; unattended probing. **Plan (owner):** interviews and ticket coding by M2, week 8 (Lead 1); by M3, week 11, the boards, T7 and one product's low-voltage section transcribed and checked against its manual (Lead 2) and the full evaluation (Lead 3); S1 and live runs by M5, week 13 (Lead 4). Late results stay pending, never replaced.

DESIGN DECISION · Score by ambiguity group and effort to a confirmed answer

Options: per-part accuracy and a count of measurements, or group accuracy and weighted effort to a confirmed answer.

Choice: the latter: per-part scoring rewards lucky guesses between faults with identical readings, and an answer is worth acting on only once confirmed. **Consequence:** group sizes and raw effort are reported too.

Part 4 — Preliminary AI approach, risks and ethics

Section lead: MEMBER_4

Why a hybrid. Language models guess when uncertain, partly because training and evaluation reward guessing [54]. Alone, a model has no expected readings for this unit under tolerance, no calibrated belief over hundreds of suspects and no cost per measurement; with a simulator as a tool [19] it checks one guess at a time but lacks the last two (T5 and a model-with-simulator baseline test this). Simulation supplies the physics; Bayesian inference keeps a probability for every suspect group, updated after each reading through likelihoods learned from simulated units: one to three full-covariance Gaussians per fault (chosen by the Bayesian information criterion) over the circuit's readings, so any subset of readings has a closed-form likelihood. "No single fault fits" is a hypothesis with a broad flat density, set on as-new validation units to flag at most 5% of single faults. Suspects start equal (the field's fault mix is unknown); the complaint's symptoms reweight them through a symptom model learned from simulation; T14 tests a wrong prior. Boosted trees [55] are a comparison engine (as-new pilot top-1 without the complaint: 91.8% against 94.3%). The next measurement maximizes expected information gain per unit of effort, the one-step entropy rule of model-based diagnosis [56] with a cost term (an optimal sequence needs a search [57]); a diagnosis stops at 90% belief in one group, at the budget, or when nothing informative remains. The language model (Claude, version pinned) reads complaints (offline, rules do), proposes meter-photo readings for confirmation (offline, image processing) and explains; every number it writes must trace to a reading (T17), and it stays only if it beats a model alone (T5) and reads complaints and photos well (T18, T19).

Table 4. Risk register: likelihood/impact (L, M, H; F: fatal) before and after the control, our judgment, reviewed each milestone; Lead n signs Part n . A fatal risk needs a code lock or stays out of scope; NIST’s AI RMF is a checklist, not a claim of conformance [72].

Risk	Before	Control	Check	Owner	After
Shock during a guided step	M/F	Hazard map; bring-up, discharge locks from forms only	T15, T16, T24	Lead 4	L/F
Unqualified user	M/F	Businesses only; terms of use; named supervisor for trainees	T25	Lead 4	M/F
Unit differs from its model	M/F	Checked against the chart, else every point hands-off	T13, T15	Lead 2	L/F
Owner’s unit damaged	M/M	Owner approves removals; leads clipped on unpowered	T13, S1	Lead 3	L/M
Invented reading; injection	M/H	Number check; photos confirmed; text as data; model pinned	T16, T17	Lead 4	L/M
Simulation-to-reality gap	H/H	Meter load, mains, aging simulated; fault audit; real boards	T7	Lead 2	M/H
Over-trust; deskilling; blame	M/H	Reasons, runners-up shown; trainee mode; sign-off limited	T25, S1	Lead 4	M/M
Tickets used against staff	M/M	Roles, not names; kept by the shop; forbidden by the terms	Ticket test	Lead 1	L/M
Misleading performance claims	M/H	Only locked results quoted, labeled simulated or real	Claims check	Lead 3	L/M
Privacy of complaints, photos	M/M	Offline default; metadata stripped; session-only data	Metadata test	Lead 4	L/M
Copied manuals, schematics	M/M	Shops model manuals they hold; none shipped	File audit	Lead 1	L/M
Boards, review or API key late	M/M	Pending results reported with the command	Milestones	Lead 3	L/M

Set aside: consistency-based candidates [56] (tolerances blur conflicts); retrieval over manuals (they describe healthy units and common faults); deep networks that condition on any subset of readings [58, 59] and choose measurements by information gain [60], but harder to audit than one density per fault. **Autonomy:** on five levels adapted from [61] (L0, none, to L4, unattended), Differential is L2: it ranks suspects and recommends steps; a person takes every reading.

Safety. The hazard map covers healthy and faulted units, and a fault can put high voltage where there is normally none: channel-strip point TP10, normally about 0 V, reaches 126 V under “C203 short circuit”. Steps at points above 50 V carry hands-off and discharge warnings; in a high-voltage chassis every powered step says “high voltage inside”, names a meter rated CAT II 600 V or better and waits until the technician records in a form how the unit was brought up (variac, series lamp, current-limited supply, or already running). Before a part is removed, the owner’s approval and a reading under 2 V at every point that can hold high voltage are recorded; the check lapses after 5 minutes unless a bleeder is clipped across the main filter capacitor, as a discharged capacitor recovers charge [35], and on any power-up. These are checks, not guarantees: the locks act on the technician’s form entries, so they bound what the tool recommends and records, not what a person does; the residual shock risk rests on the user’s unchecked qualification, and the hazard map covers only the modeled circuit. It cannot yet recommend the unpowered checks bench practice starts with (resistance, diode, ESR).

People. Runner-up suspects and reasons are shown, yet AI can mislead: consultants given GPT-4 were 19 percentage points less likely to solve a task outside its competence [62], and those trusting AI more report less critical thinking [63]. Trainees commit their own next step before seeing the tool’s and need a named supervisor near high voltage; a tutor giving hints instead of answers largely mitigated the learning loss an unrestricted chat model caused [64]. S1 plants one wrong recommendation on a non-hazardous step of a 24 V board, after the IRB’s determination; participants are told beforehand that some advice may be wrong, are debriefed and recruited outside their employers (T25). Whether owners turn faster diagnosis into a faster pace is a question for the interviews. As fairness audits do for groups of people [65], errors are broken down by complaint voice, since owners who describe faults in plain words could get worse diagnoses (non-English complaints are not covered; breakdowns by fault type show testability, not fairness): a voice more than 5 points worse on the T1 or T6 units, with an interval excluding zero, blocks release (T26).

Accountability. The team answers for the hazard map, stated performance and corrections; a shop, for its circuit models and staff; the technician, for measurements and precautions. Responsibility should follow control, lest the technician become a moral crumple zone, blamed for a system they cannot steer [66]: the repair ticket records the tool’s version, what it recommended and why, and what the technician entered in the forms; it stays in the shop’s system, and the terms of use forbid using it to rate, discipline or pay staff. Of the rules in Part 1, OSHA’s is met by selling to businesses and supervising trainees, the AI Act’s by telling owners of the AI and forbidding staff rating, and the FTC’s by quoting only locked results, labeled simulated or real. Whether advice software is a product (strict liability) or a service (negligence) is unsettled in the U.S.: bills would make AI systems products [67, 68] or shield developers whose tools licensed professionals use, which bench technicians are not [69]; the revised EU directive makes software placed on the market after 9 December 2026 a product [70], and California bars defending a harm claim by saying the AI acted autonomously [71].

DESIGN DECISION · Safety and security rules live in code, not only in the prompt

Options: prompt the model to be safe, or enforce safety in code. **Choice:** code: prompt injection ranks first among LLM application risks [73] and injected text can steer a model [74], so notes, complaints and photo text are data, the model’s tools act only on the session and safety records come only from the forms. **Consequence:** the same warnings, refusals and locks in every mode.

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